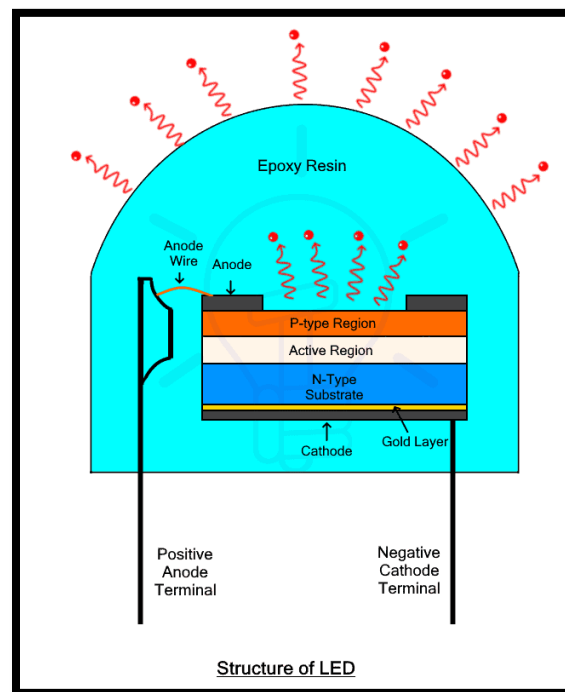


## Lab 2: Light Emitting Diodes

Leo Marcinkus

Lab Partner: Jimmy Shapiro

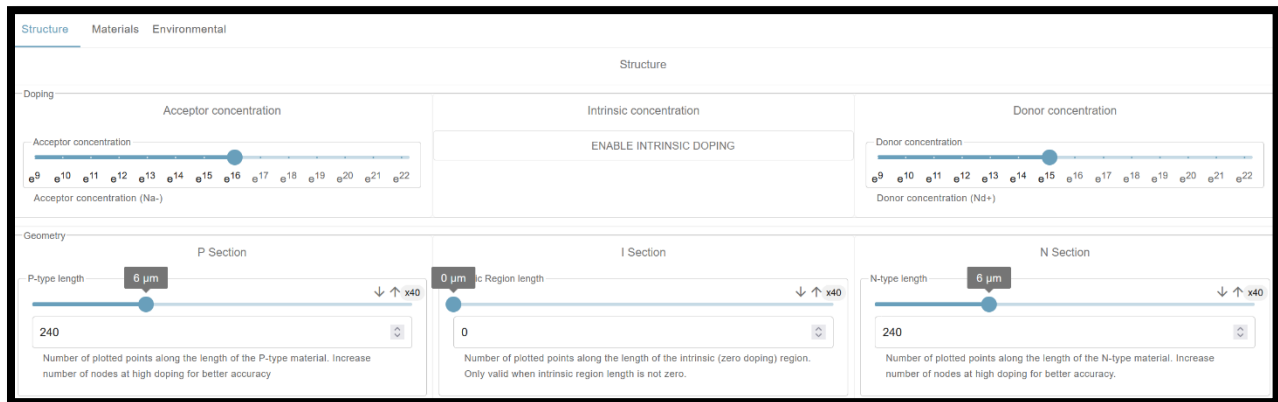
This lab investigates the behaviors of standard PN junction light emitting diodes. They are small semiconductors that only emit light when forward biased. With an applied voltage, electrons and holes move to the junction and emit light. The color depends on the material of the LED. They are used in a massive number of electronic devices as they are power efficient and produce little heat. They are manufactured in a variety of sizes allowing for a large amount of versatility. Their easily adjusted brightness, low maintenance, and multitude of color options make them a valuable component. The basic I-V properties, brightness-voltage, and power behaviors of these devices are studied in this lab.



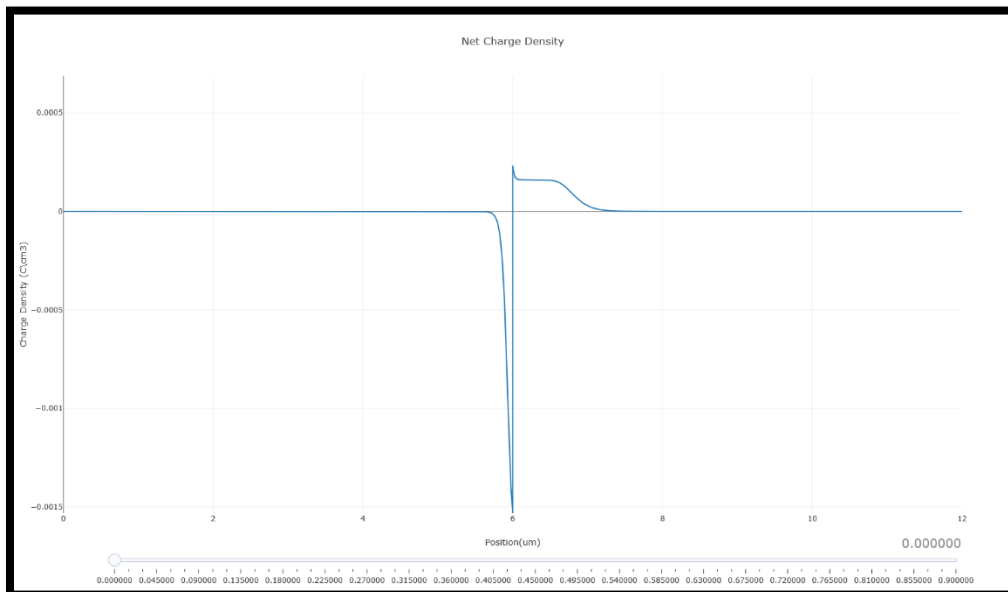
*Figure 1: Light Emitting Diode Structure*

## Fundamentals:

The nanoHUB PN Junction simulation was run for two different doping densities. The acceptor concentration was raised to  $e^{16}$  first for the simulations. Then the donor concentration was raised to  $e^{16}$  after returning the acceptor concentration to its default  $e^{15}$  value. The plots are as follows:



*Figure 2: Acceptor Concentration Increased to  $e^{16}$*



*Figure 3: Charge Density when Acceptor Concentration Increased to  $e^{16}$*

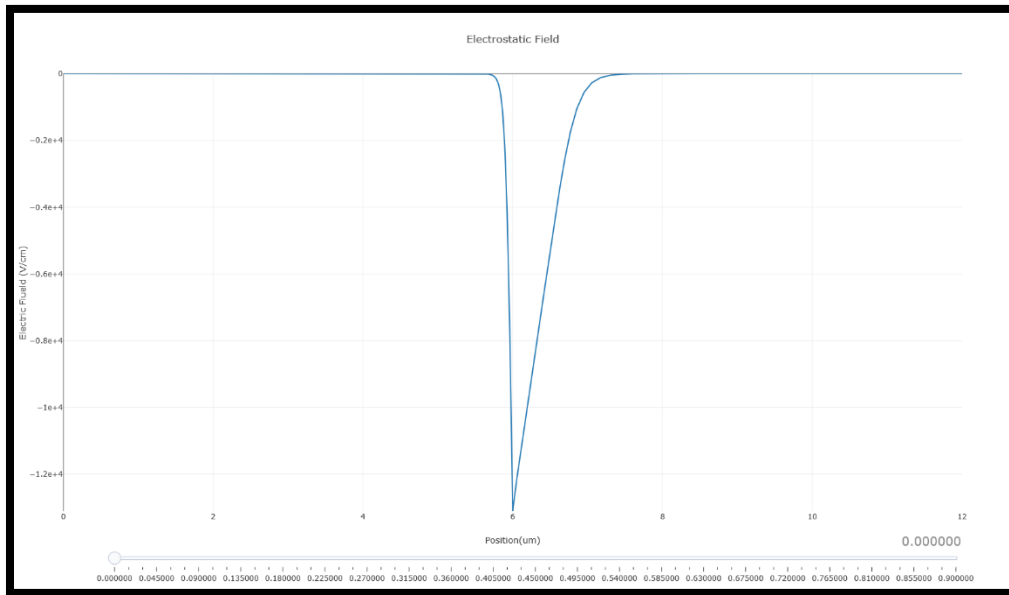
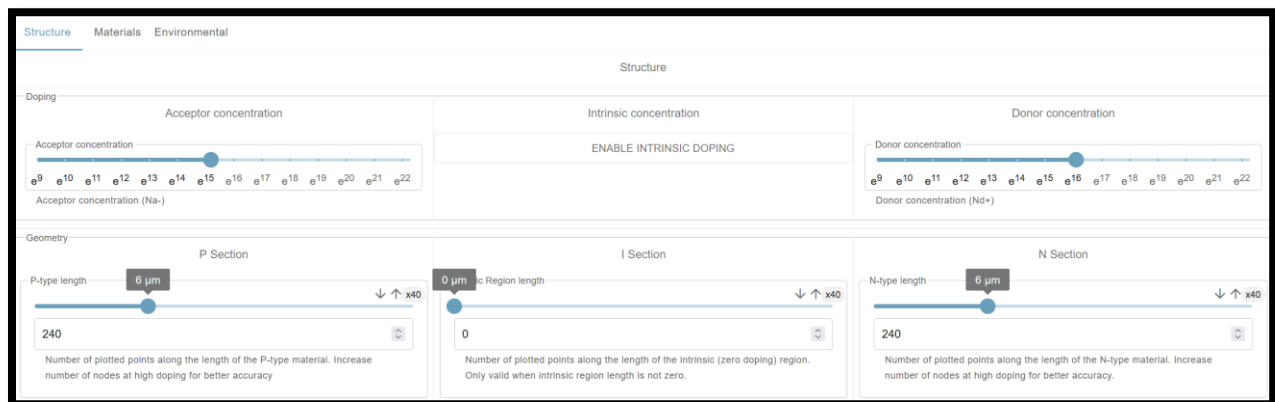


Figure 4: Electric Field when Acceptor Concentration Increased to  $e^{16}$  Figure 5: Donor



Concentration Increased to  $e^{16}$

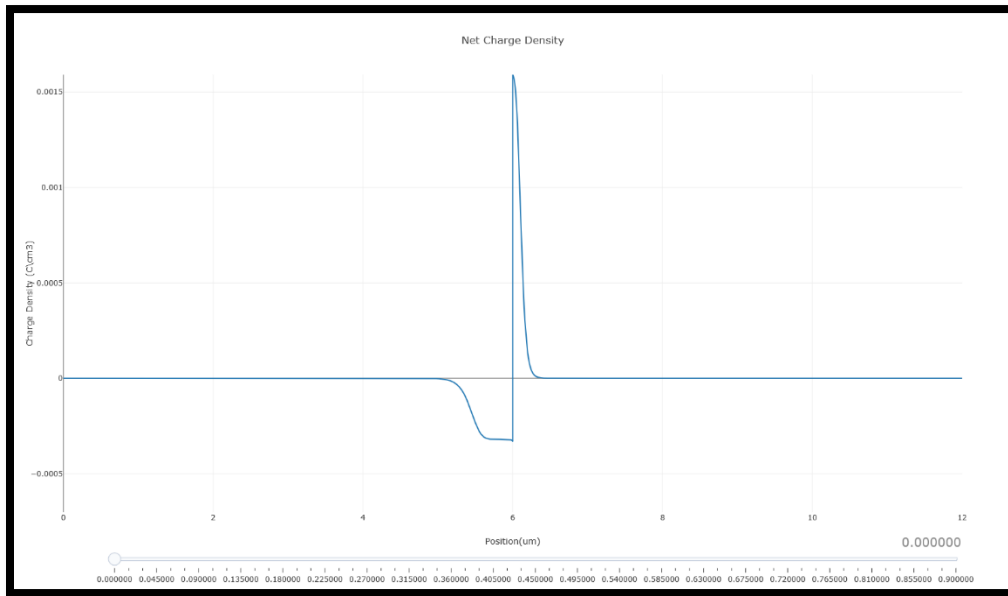


Figure 6: Charge Density when Donor Concentration Increased to  $e^{16}$

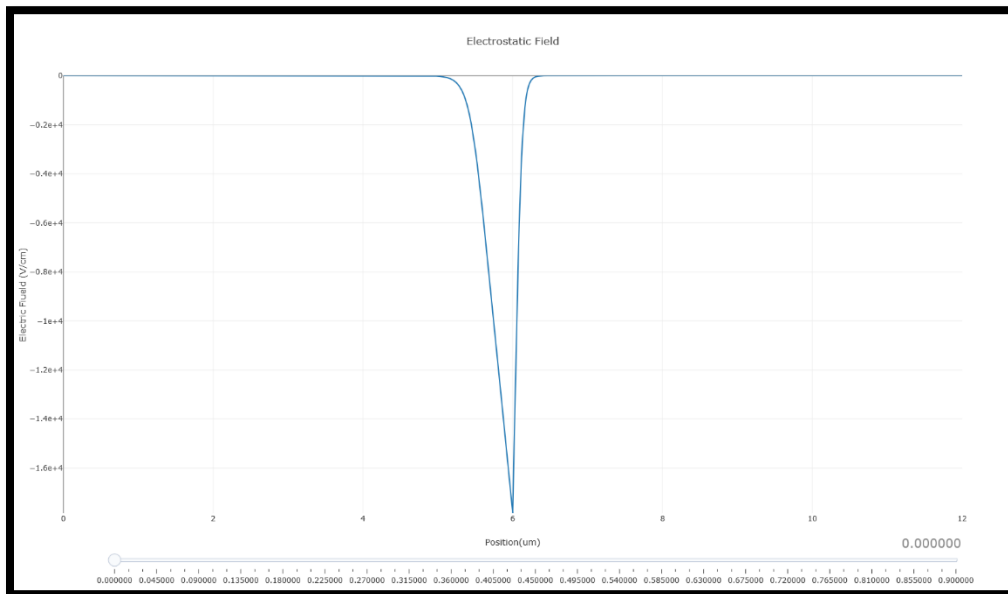


Figure 7: Electric Field when Donor Concentration Increased to  $e^{16}$

The simulation provides expected results. When only the acceptor concentration is raised, the negative peak in the charge density increases. If only the donor concentration is increased,

then the positive peak in the charge density increases. When the acceptor concentration is doped, the negative peak should be on the left side of the electric field graph. It is the opposite for when the donor concentration is doped. The peak should be on the right side of the plot.

#### Function of an LED:

Similar to what was described in the device behavior, an LED works by forward biasing a PN junction so that electrons from the n-type side and holes from the p-type side recombine across the band gap. When this combination occurs, energy is released as photons, with the photon wavelength set by the semiconductor band gap. This is what allows for the LED color to be altered based on material. LED efficiency depends on two things. The first, how many electron-hole pairs recombine radiatively versus non-radiatively. More simplified, radiatively meaning the electron-hole pair ends up producing light in the form of a photon and non-radiatively meaning that pair produces heat instead. The second thing the efficiency depends on is how many photons escape the device without being reabsorbed. These determine the power efficiency of the LED and its color.

#### Converting Photon Wavelength to Energy:

$$E = \frac{hc}{\lambda}$$

$$h = 6.62607015 \times 10^{-34} \text{ J} \cdot \text{s}$$

$$c = 2.99792458 \times 10^8 \text{ m/s}$$

$$1 \text{ eV} = 1.602176634 \times 10^{-19} \text{ J}$$

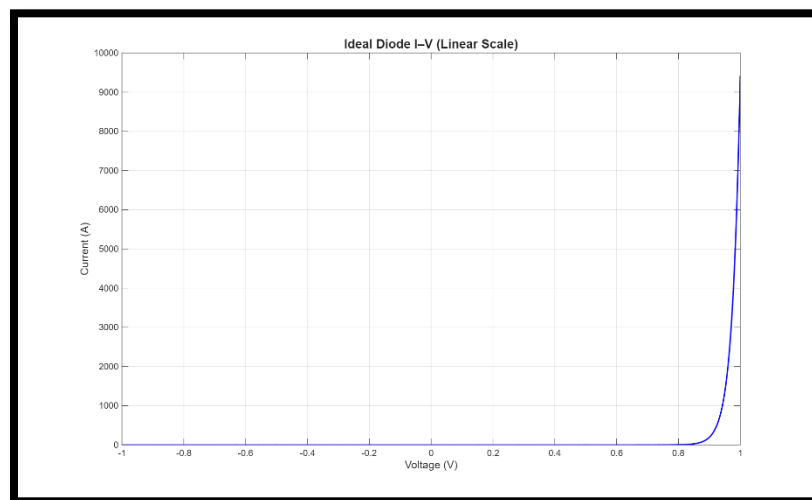
$$\frac{hc}{1 \text{ eV}} = \frac{1.98645 \times 10^{-34-25} \text{ J} \cdot \text{m}}{1.602176634 \times 10^{-19} \text{ J}} = 1.23984 \times 10^{-6} \text{ eV} \cdot \text{m} = \frac{1240 \text{ eV} \cdot \text{nm}}{\lambda \text{ in nm}} \approx E(\text{eV})$$

## The I-V Relationship of a PN Junction: The Ideal Diode Equation

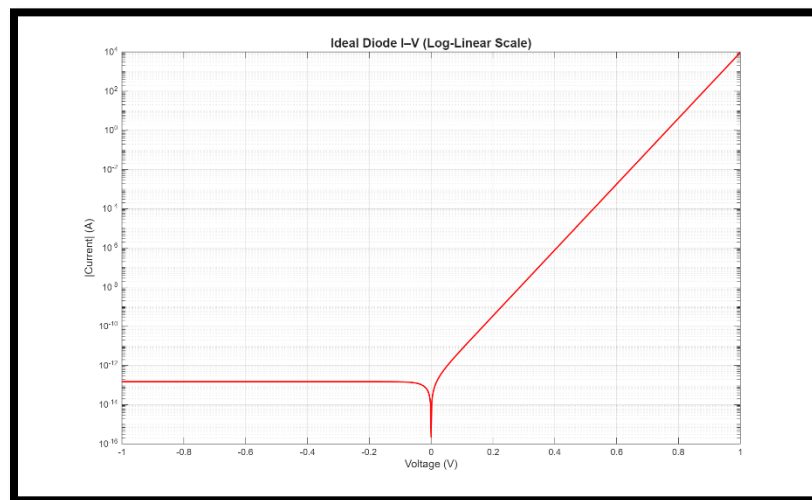
The values and plots were calculated utilizing MATLAB.

```
>> FundamentalsPartD
kT/q = 0.025852 V
A = 0.001 cm^2
Dn = 34.7709 cm^2/s
Dp = 11.8402 cm^2/s
Ln = 0.00589669 cm (58.97 um)
Lp = 0.00344096 cm (34.41 um)
I0 = 1.49606e-13 A
```

*Figure 8: Calculated Values*



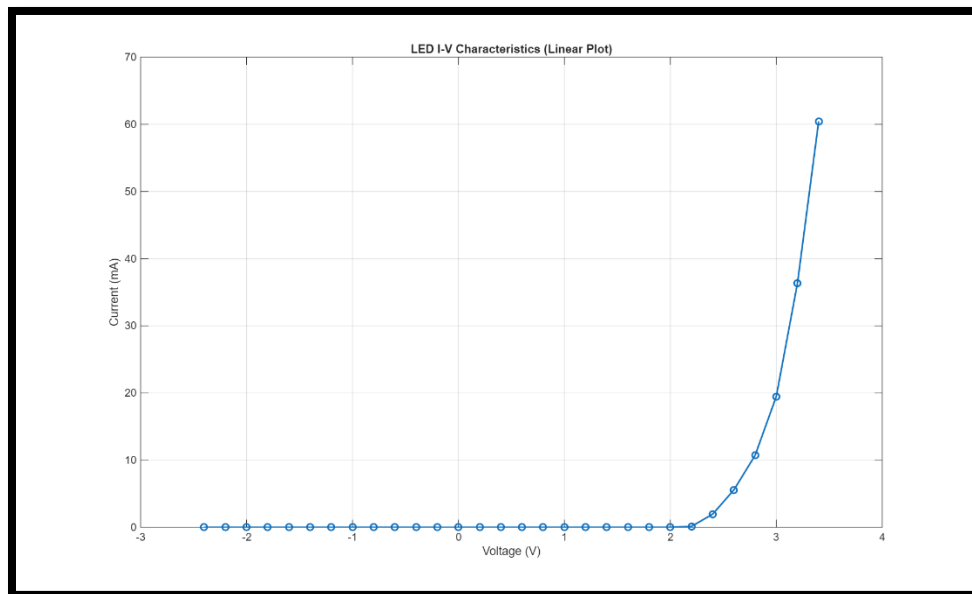
*Figure 9: Linear Plot*



*Figure 10: Log-Linear Plot*

### Experiment 1: Characterizing an LED's I-V Properties

The goal of this experiment is to understand the limits of the LED and observe what the values look like when plotted. This was done by taking current measurements from -2.4 V to 3.4 V with a green LED. Data was only recorded up to 3.4 V because any higher would exceed the 200mW power rating of the LED. The setup was simple with a standard power supply and digital multimeter. The I-V curve is plotted with a linear and semi-log graph made by MATLAB.



*Figure 11: Green LED I-V Curve Linear Plot*

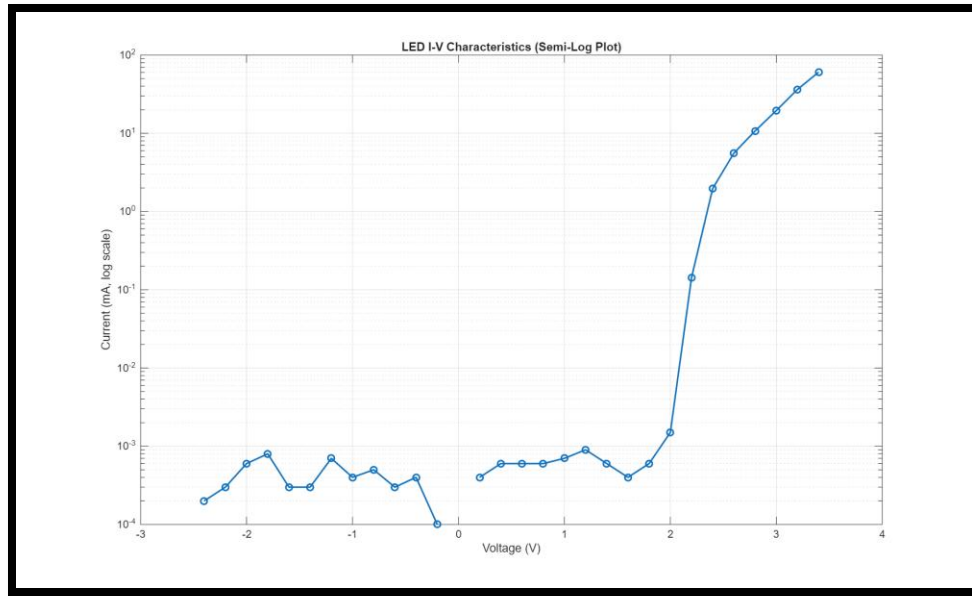
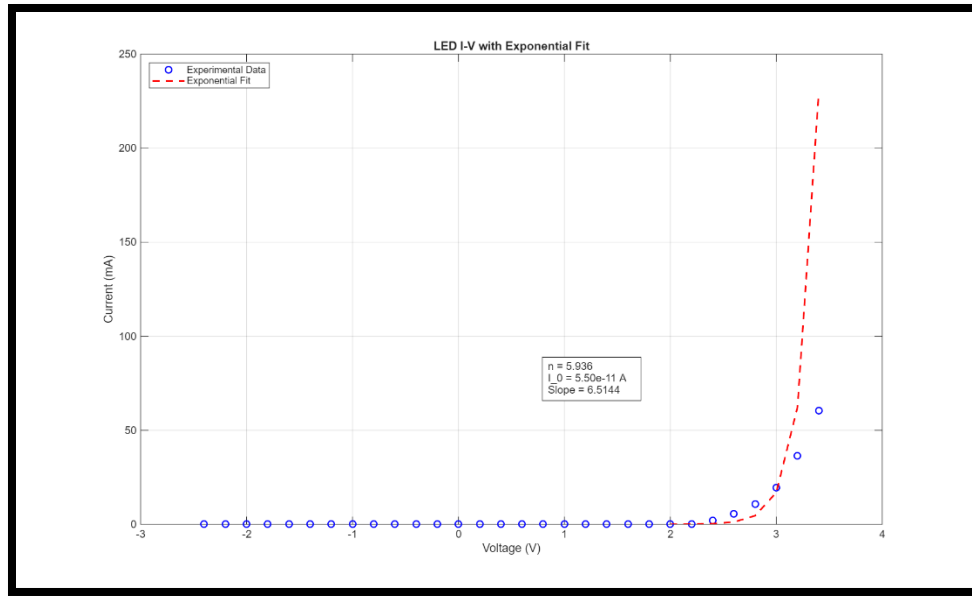


Figure 12: Green LED I-V Curve Semilog Plot

Both plots behave as expected. The odd behavior on the semilog plot can be attributed to internal resistance of the power supply and the digital multimeter. For lower voltages, no values were being accurately measured as they were out of the scope of the instruments. Placeholders were still recorded instead of “0 mA” so it could be noted when the LED behavior changed. The I-V curve data is now fit to the *non*-ideal diode equation. Once again utilizing MATLAB.



*Figure 13: Green LED I-V Curve Fit to Non-Ideal Diode Equation*

The  $n$  value is calculated to be 5.936 with  $I_0$  being 6.5144 A. These values are extraordinarily high but that is because the fitting window includes the whole range of voltages. If the window was shortened to when the current measurements started to increase (2.0 V to 3.4 V), the calculated values for  $n$  and  $I_0$  would dramatically decrease. But if the whole range of -2.4 V to 3.4 V is utilized, those calculated values will remain large to ensure a better fit. It was difficult to measure, but the  $I_0$  value for the green LED appears to be around 1  $\mu\text{A}$ .

The linear plot shows the overall I-V behavior of the LED. Current is almost zero until a large increase when the LED reaches its turn on voltage. The semilog plot makes it look a little different, however. The exponential look behavior of the linear plot now looks linear in the semilog plot. There are more discrepancies visualized in the semilog plot as the y-axis is plotted differently. Small inconsistencies are magnified making deviations from what is expected more obvious. The data fits the non-ideal diode equation somewhat accurately. It should be more exponential but with higher voltages this should line up. However, the chosen values for  $n$  and  $I_0$

are extremely large in order to make this happen. If using a directly calculated value instead of a free parameter, it would drastically change. Compared to the regular diodes analyzed in lab one, this LED does not match up as well to the same non-ideal curve. This is expected as an LED is a more complicated diode than a standard small signal diode like previously measured. Meaning there more reasons for measurements to be slightly off (LEDs usually a large  $n$  value). The way the experiment was wired, series resistance, and temperature increase can help explain discrepancies. Series resistance does not do much until larger currents are drawn. But when it is causing a change, it will usually drag an I-V curve down by slowing exponential growth. The  $I_0$  used in the I-V fit doesn't come anywhere close to the directly measured value. It is significantly higher than  $1\ \mu\text{A}$ . If a better voltage range was selected, the calculated  $I_0$  value would be closer to the measured  $I_0$ , but would still be off. Reasons for this difference can include current leakage, precision of instruments, and the fitted voltage range. This can be fixed by taking tighter data points and limit the voltage range used when measuring the LED's properties.

## Experiment 2: Characterizing the Current-Voltage and Brightness-Voltage Properties of Different LEDs

Experiment 2's goal is to observe the different behaviors between three different colored LEDs. The colors chosen were red, green, and blue. A photodiode was used to be able to measure these differences. The setup was similar to experiment one. A power supply and digital multimeter were the major tools utilized. The chosen LED was bent directly over the photodiode so it could provide more consistent results with each colored diode substituted in. A cardboard box was placed over this setup to ensure no external light could affect the photodiode. Each colored diode went through a different range of voltages. Each diode started at 0 V but the maximum value it went up too depended on the power rating of 200 mW. The red LED hit this limit the quickest at 2.4 V, but the green LED went all the way to 4.6 V as seen in the plots below.

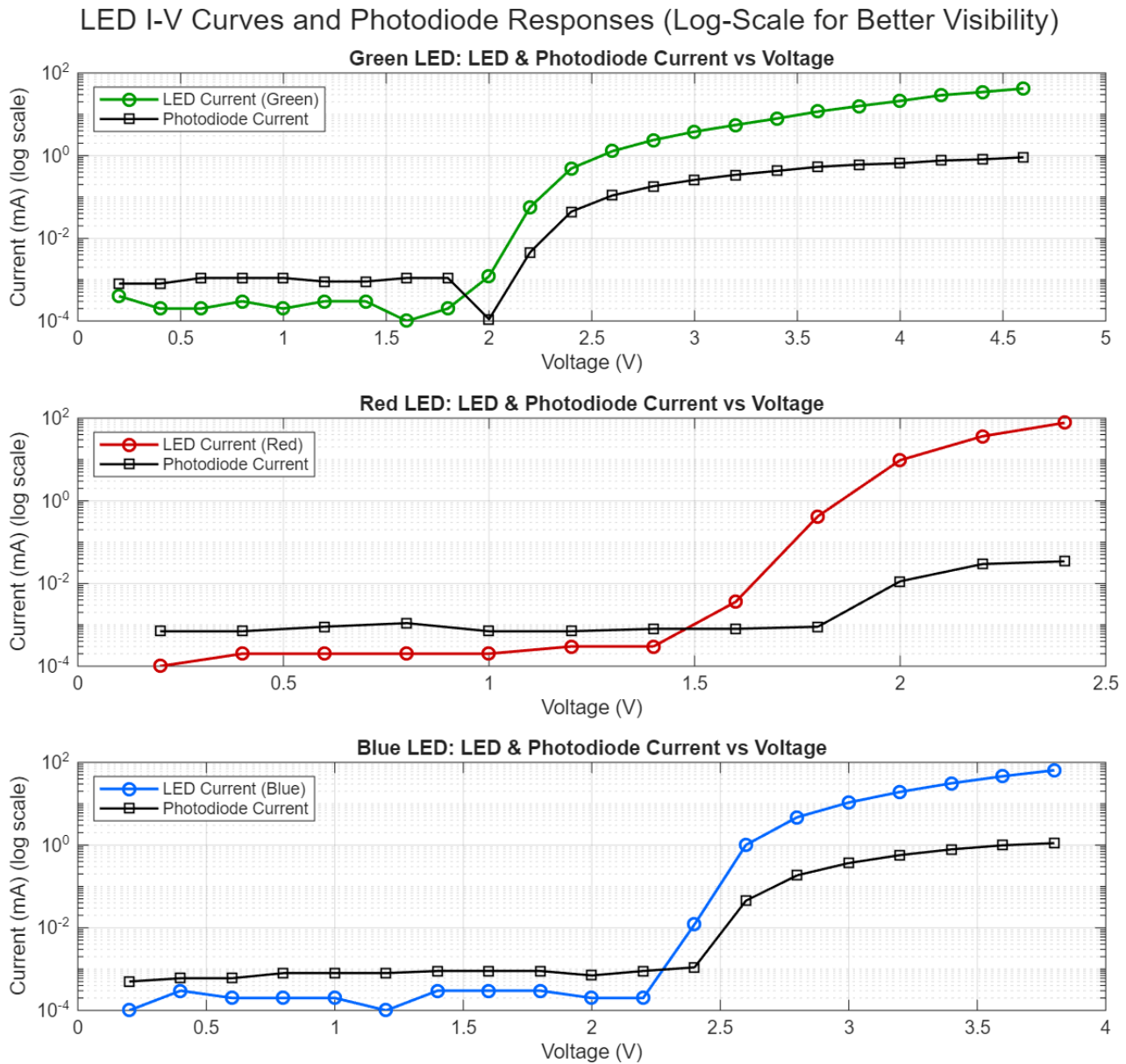


Figure 14: Green, Red, and Blue LED I-V Curves and Photodiode Current

The LED's current is overlayed with the photodiode's current, so it is easy to see the effect the LED has on the photodiode. Each LED went up to a different voltage value, but the horizontal scale remains the same for each. The y-axis is in log, so it is easier to see how the photodiode's current lines up with the LED's. In a linear plot, the photodiode's measurements would be dominated by the LEDs. The relationship is apparent. The brighter the LED gets, the more current the photodiode provides.

The relative quantum efficiency and relative power efficiency versus voltage is plotted below. The quantum efficiency is essentially how efficiently each electron turns into a photon to produce light. Relative power efficiency is as it sounds, power in versus power out. Each was normalized so they can be compared on the same plot as shown. The values were calculated using the following:

$$\text{Relative Quantum Efficiency} = \frac{I_{PD}(V)}{I_{LED}(V)}$$

$$\text{Relative Power Efficiency} = \frac{I_{PD}(V) \cdot E_{Photon}}{I_{LED}(V) \cdot V}$$

$$E_{Photon} = \frac{hc}{\lambda_{Peak}}$$

$$h = 6.62607015 \times 10^{-34} \text{ J} \cdot \text{s}$$

$$c = 2.99792458 \times 10^8 \text{ m/s}$$

$$\lambda_{Green-Peak} = 635 \text{ nm}$$

$$\lambda_{Red-Peak} = 565 \text{ nm}$$

$$\lambda_{Blue-Peak} = 470 \text{ nm}$$

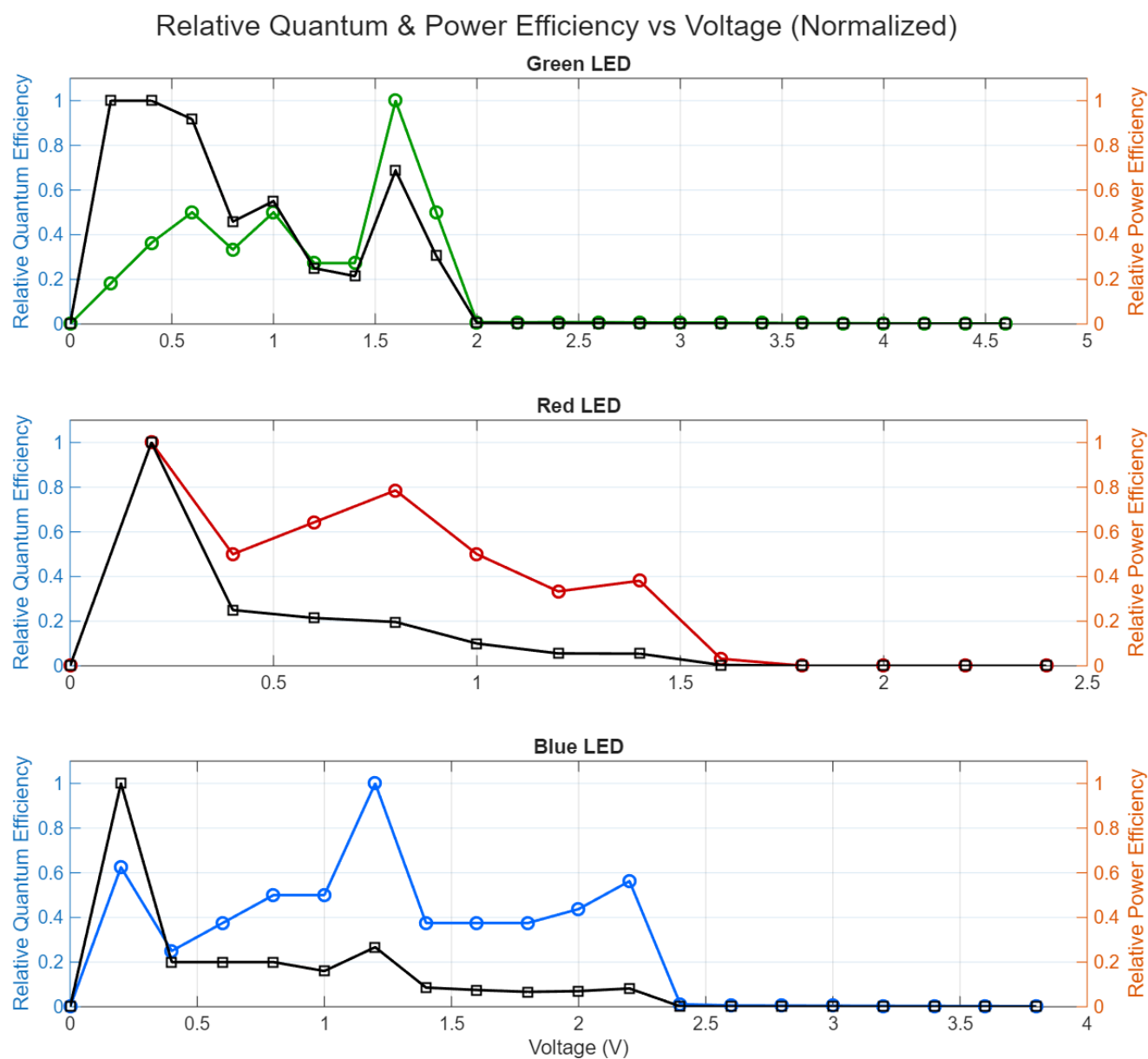
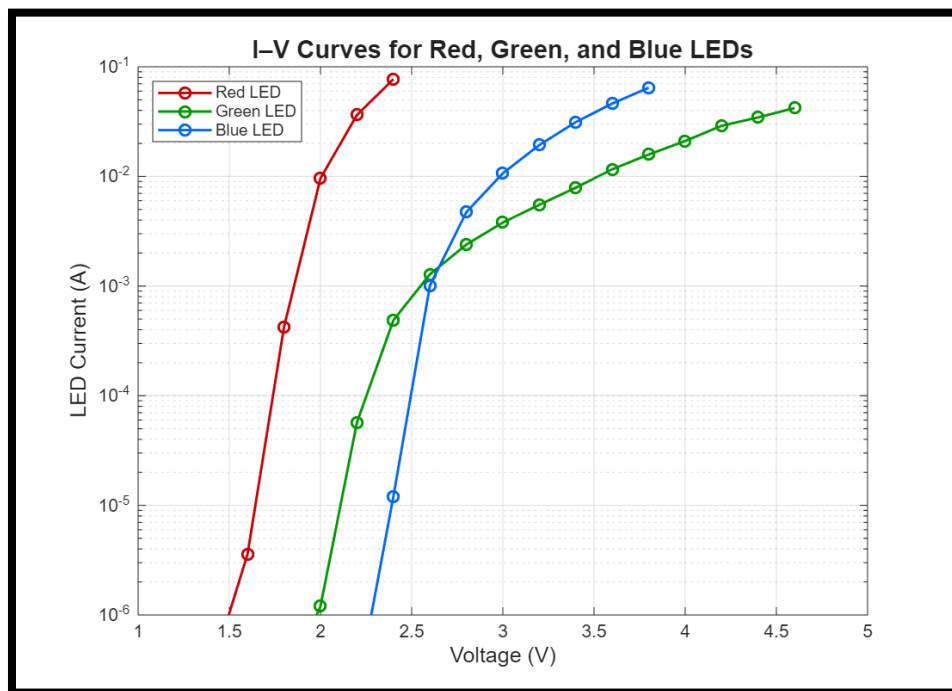


Figure 14: Green, Red, and Blue Relative Quantum Efficiency (colored) and Power Efficiency (black)

The relative quantum efficiency increases after the LED reaches its minimum voltage. After that, it starts to taper reaching an equilibrium point. The relative power efficiency appear to peak around the turn-on voltage of each LED and then start to decrease and match the quantum efficiency. The large peaks in power efficiency in the very beginning of each colored LED is not necessarily where it is most power efficient. Just an artifact of low values and not perfectly precise measurements. The most efficient operating points appear to be about 1.6 V for the green LED, about 0.75 V for the Red LED, and about 1.2 V for the blue LED.

The I-V curves for each LED can be placed on the same plot to observe the behavior between the different colors.



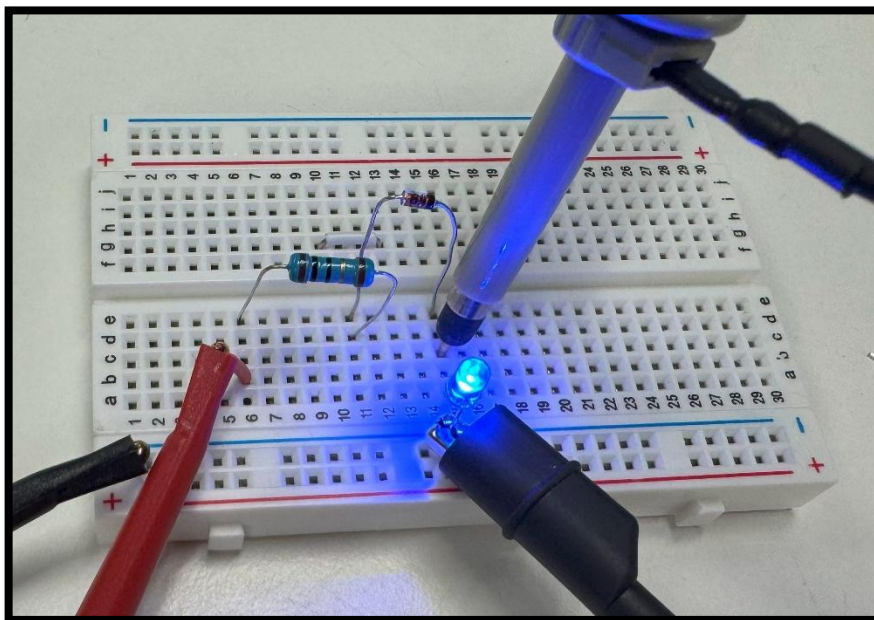
*Figure 15: Green, Red, and Blue LED I-V Curves Overlapped*

Each LED shares the similarity of there being a larger current corresponding to a larger voltage. However, each LED hit the 200mW power rating at a different voltage. The red reached the limit first at 2.4 V, the blue next at 3.8 V, and the green last at 4.6 V. The turn on voltage for

each LED is different too. The red LED has the lowest, followed by green, and blue being the highest. This lines up with the different wavelengths of each color. Red has the longest wavelength and the smallest bandgap requiring a low turn on voltage. While blue has the shortest wavelength and largest bandgap requiring a higher turn on voltage. In the power efficiency versus voltage plots, the power efficiency curves peak just above the turn on voltage. This is where the most light per watt is produced. At the voltages, energy loss to heat and light emission increases, causing efficiency to drop towards zero.

### Experiment 3 Measuring the Carrier Lifetime in LEDs Based on their Transient Response:

Experiment 3 serves to take an LED and find its carrier lifetime,  $\tau$ . The setup involves building a circuit and using a waveform generator to create a wave and an oscilloscope to measure  $dV/dt$  for the chosen LED. A  $100\ \Omega$  resistor is used in series with a diode and LED. The circuit setup is shown below:



*Figure 16: Experiment 3 Method b Circuit*

The following waveform was generated and sent through the circuit as seen through the red and black probes. The circuit output was measured and had its linear  $dV / dt$  approximated.

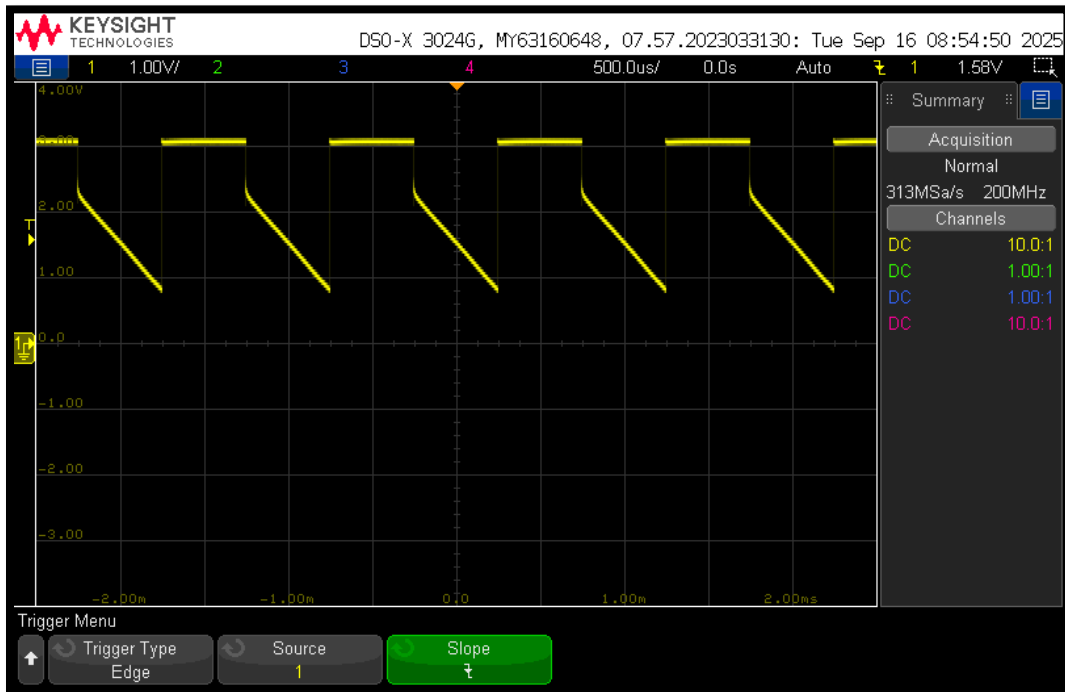


Figure 17: Waveform Generated

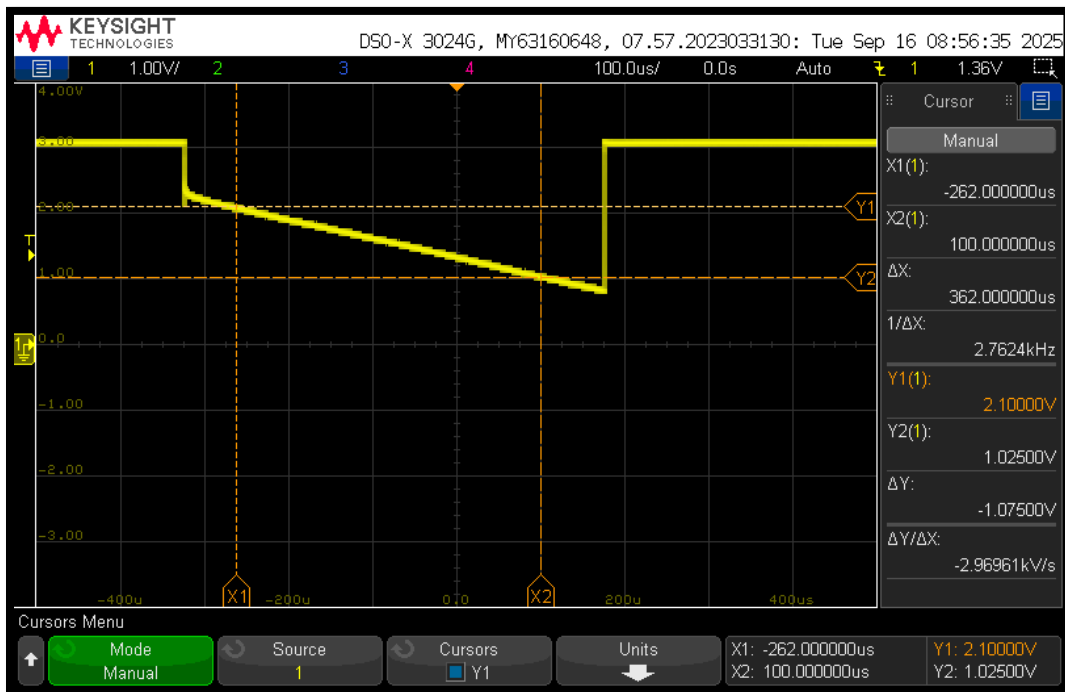


Figure 18: Output Wave Linear Measurement

The measured output  $dV / dt$  is approximately  $-1.075 \text{ V} / 362 \mu\text{s}$ . This can be put into the following formula alongside the blue LED's calculated  $n$  value to solve for  $\tau$ . The  $n$  value was calculated for a blue LED from the data recorded in experiment 2.

$$\tau = \frac{nk_bT}{q} \cdot \frac{1}{\frac{-dV}{dt}} = 39.4 \mu\text{s}$$

$$n \approx 4.52$$

$$\frac{k_bT}{q} \approx 0.02585 \text{ V (assuming } T = 300 \text{ K)}$$

$$\frac{-dV}{dt} = -1.075 \text{ V} / 362 \mu\text{s}$$

The  $n$  value being used is quite large which distorts the calculated carrier lifetime. I believe this to be because the blue LED used in experiment 2 is far from ideal. The linear voltage range used to calculate the value could have been more optimized if more data points were recorded. Next, the maximum response frequency of my eye was tested by reducing the frequency of the square wave input. The lowest frequency where I could still discern the blinking of the LED was about 30 Hz.

The oscilloscope clearly shows a linear decay in voltage after each pulse. There are small artifacts before and after but those can be attributed to noise and on/off effects of the resistor and diode. Overall, the shape and behavior lines up well with the shape in the paper by Dalapati, et al. With the calculated carrier lifetime to be  $39.4 \mu\text{s}$ , the maximum frequency that can drive the LED can be calculated as such:

$$f_{\text{maximum}} = \frac{1}{2\pi\tau} = \frac{1}{2\pi(39.4\mu\text{s})} \approx 4.04 \text{ kHz}$$

This frequency is suitable for simple but useful tasks that LEDs are already commonly used for. Something like indicators, signal pulses, and basic light functions. However, it is nowhere near fast enough for something like high-speed communication and data delivery. Something like fiber optic cable, as an example. The implications of being able to see the light flicker at 30 Hz shows that the human eye doesn't necessarily need something incredibly fast to appear to be sitting still. It is still beneficial for an LED to pulse at a much faster rate than just above the detection of the average human eye, however. For technologies such as screens, a much higher frequency is needed for it to appear smooth. You might have heard of the "refresh rate" of a screen before. The higher the frequency the more beneficial. Any action on a high refresh rate screen, such as 165 Hz, will appear more fluid and natural. But if a screen could only handle a 30 Hz refresh rate, using it would feel sluggish and stutter because the human eye could detect each change.

### Conclusion:

The general behavior of LEDs and how their color can affect their properties was analyzed. I-V curves and efficiencies were measured for multiple colors to differentiate their properties. I learned that, in general, an LED still behaves as a regular diode. It has a forward and reverse bias with a turn on voltage. Efficiency and brightness of these LEDs vary based on their color's wavelength. Exploring carrier lifetime in experiment three showed me how the device an LED is utilized in acts as its primary constraint. My largest takeaway is that it is important to choose a correct scale and voltage range to convey data properly. When recording data on a small scale like this, every component can cause an issue and distort measurements. I learned useful knowledge of the LED and its behavior in this lab and how they function in modern devices.

Generative AI Usage:

AI was used to help level out my plots and scale them correctly. ChatGPT was the engine used for this as it is quick to help fix my MATLAB code. The prompts utilized were simple. I would provide my recorded data and code I created to generate plots then would ask something like “Adjust and edit the scale of the y-axis to ensure both curves can be clearly read.” It always had an issue setting the colors of each curve. But this is a simple issue I can correct myself. I found it useful to check if my recorded data was realistic as well. It helped sort out discrepancies and reveal odd behavior that should be remeasured.

Citations:

Introduction LED structure image:

"Structure of LED (Light Emitting Diode)." *Electrical Technology*, 10 June 2022,

[www.electricaltechnology.org/2022/06/led-light-emitting-diode.html](http://www.electricaltechnology.org/2022/06/led-light-emitting-diode.html)